



Haptic Feedback Remote Control System for Electric Mechanical Assembly Vehicle Developed to Avoid Obstacles

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Abstract

In recent decades, various solutions have been proposed for the navigation of vehicles in unknown environments to avoid obstacles in order to perform safer driving. In this article, a designed electric mechanical assembly vehicle provided by Mecanum 4-omnidirectional wheels holonomic on the corners of a rectangular base has been investigated with the ability to detect obstacles by combining the developed algorithms and sensing system. The vehicle is equipped with LiDAR and communication is guaranteed by WiFi. The interface with the mechanical and electrical devices of four wheeled Mecanum vehicle is well-executed by haptic feedback remote control using an analog joystick that guarantees the driving movements. Experimental tests were conducted to avoid obstacles and detect collision by the electric vehicle, in case of hazard detection. In this investigation, the obstacle is placed at the distance of 2 m away from the vehicle. If the resistance decreases, it means that the vehicle is approaching an obstacle. The recorded speed of vehicle passes from 350 mm/s to less than 10 mm/s, on the contrast, the resistance increases to reach 100%.

Keywords Haptic feedback · Obstacle avoidance · Vehicle safety system

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1 Introduction

In modern technology, haptics and tactile sensation refer to the interaction between human touch and a computer interface attempting to emulate acoustic, physical, or visual stimuli. Haptic feedback and controllers provide real-time environmental information to the drivers by equipping modern vehicles with sensor systems such as radar, cameras and laser systems [1]. Predictive controllers can share control with the drivers to improve the environmental perception and safety of modern vehicles in nuclear power plants, military operations, space exploration and agriculture. The effect of tactile feedback on the teleoperation system and the performance of an unmanned remote controlled ground vehicle are analyzed for their stability, performance, operator control efforts in [2, 3]. Obstacle avoidance is the basis for the navigation of a highly exposed to collisions, material damage or human safety vehicle [4]. A general framework is applied to design tactile remote operation algorithms in order to command robotic aerial vehicles by using local nonlinear control loop. The mechanical model used for local control provides the pilot with intuitive command and feedback on the vehicle's movement and environment [5].

The haptic teleoperation of Unmanned Aerial Vehicles (UAVs) requires the remote control by a human operator based on sensor data and visual information provided by a camera on board of the vehicle in order to dissuade the driver from the dangerous path. The teleoperation interface provides force feedback based on the path generated by an obstacle avoidance algorithm [6]. UAV embedded with a haptic teleoperation system, in complex environments, reduces operator workload by applying motion and controllers. Vehicle speed is governed by the pilot via tactile feedback on master device, setting target position and reference velocity along a servo-controlled trajectory with an improvement on performance of the navigation system. This scheme guarantees a safer and faster air guidance of the multi-rotor robotic vehicle, with low workloads and cognitive levels [7]. The interaction force between environment and Miniature Aerial Vehicle (MAV) is modeled by using two different approaches as the artificial and virtual impedance force field for bilateral telemanipulation [8]. In [9, 10] the proposed collision-avoidance system has a network of near real-time embedded subsystem and it is designed to take advantage of three operating conditions to control the warning and speed of an electric vehicle when a collision is imminent. The vehicle navigation, plans and controls, in a complex environment, the optimal trajectory through the vehicle actuators to reach a specific destination in autonomy [11]. Moreover the development of intelligent transportation system (ITS) technologies, has allowed real-time traffic data to be obtained using loop detectors, video detectors, Bluetooth sensors or radar sensors. In addition this type of technology can also be used in different robots and in healthcare [12, 13]. The dynamic obstacle avoidance architecture is based on a sigmoid function to avoid detected obstacle, and the control guidance employs this obstacle avoidance trajectory to generate the proper steering angle [14–17]. The concept of velocity obstacles which characterizes the set of trajectories that lead to a collision between interacting agents is investigated in [18] using three algorithms for local motion planning as a centralized convex optimization in which a joint quadratic cost function is minimized subject to linear and quadratic constraints, a distributed convex optimization, and a centralized non-convex optimization with binary variables in which the global optimum can be found. An improvement to the cross-track error calculation mechanism in unmanned aerial vehicles for collision avoidance mechanism is made by employing the extended Kalman filter (EKF) to estimate the cross-track error in [19, 20]. Fast Geometric Avoidance algorithm (FGA), combines geometric avoidance of obstacles and selection of a critical avoidance start time based on kinematic considerations, collision likelihood, and navigation constraints is presented in [21, 22] In order to enhance the area coverage of such

UAV swarms, a novel mobility model has been presented combining an Ant Colony algorithm with chaotic dynamics (CACOC) [23]. The electric vehicles use ultrasonic sensors, LiDAR sensors and cameras to detect obstacles in the military and transport [24, 25]. In addition this type of technology can also be used in different robots [13]. The servomotors are used extensively in industry however the power of stepper motor is low and only a miniature robot can be moved by using them [26]. A robot with servo motors and mechanical wheels can be used in healthcare: to move beds of patients in the hospital or to assist disabled person in a mobile wheelchair [12]. The dynamic obstacle avoidance architecture is accorded to an evidential occupancy grid with the path planning generated by a smooth trajectory based on a sigmoid function to avoid detected obstacle, and the control guidance employs this obstacle avoidance trajectory to generate the proper steering angle [15, 27]. Recently, electric wheelchairs with independent driving wheels modeled as two-wheeled nonholonomic vehicle models have been widely used, and avoidance of moving obstacles in narrow environment has been studied in [28].

The capabilities of autonomous mobile robots are determined by tracking, localization, navigation, and avoiding dangers and obstacles using a great variety of sensors, external references and algorithms [29–31]. Intelligent electric vehicles as self-driving and highly automated vehicles provide safer transportation with little human interaction, navigation avoiding, detecting and classifying objects in the surroundings and predicting the safest routes to take. The interactions with the environment require sensors, radars and LiDARs (Light Detection and Ranging) to determine the position and distance of an object, and the camera to detect all around the vehicle. Even multi-LiDAR sensors can improve the environmental perception of self-driving cars [32]. As explained in [33], the humans interaction with highly automated or self-driving vehicles is referred to three domains in the vehicle cabin, around the vehicle, and inside surrounding vehicles to understand, model and predict human agents. Perceptual systems such as the Multi-LiDAR sensors have the potential to improve as environmental perception systems and position estimation systems for self-driving vehicles [34]. In [35] is described the system architecture of KUVe (KIST Unmanned Vehicle Electric) that estimates and tracks the boundary of road such as curb and line using a laser range finder and a vision camera and unmanned autonomous navigation of it in KIST which is an electric light-duty vehicle equipped with two laser range finders, a vision camera, a differential GPS system, an inertial measurement unit, odometers, and control computers for autonomous navigation. A vehicle control interface that includes a tactile joystick and five 1 degree-of-freedom manipulators on the joystick, providing force sensations to the driver's hand and fingers from the

surrounding manipulators to identify obstacles that need to be predicted and avoided is proposed in [36]. The obstacle detection system for driving safety and stability, described in [37], consists of laser system components and vision system components based on the Joint Architecture for Unmanned Systems(JAUS).

In this paper, a designed electric mechanical assembly vehicle, equipped with four Mecanum wheels on the corners of a rectangular base with omnidirectional motion and able to detect obstacles by combining the developed algorithms and detection system, has been investigated. The electric vehicle provided by Mecanum 4-omnidirectional wheels is extremely maneuverable helpful in different indoor and outdoor applications and it is also able to move easily in different directions. The interface with vehicle's mechanical and electrical devices is well-executed by haptic feedback remote control using an analog joystick guaranteeing the driving movements. In the haptic feedback teleoperation, the user manipulates the vehicle or remote objects through a joystick whilst feeling a force feedback. The haptic control system includes auditory and haptic senses such as kinesthetic and tactile interface in addition to visual data to provide input information by multimodal sensory to a human teleoperator [38]. The operator receives tactile information through a force feedback joystick. The force for the joystick is calculated from the values of the strain gauge force sensor. The electrical system has involved the control of the motor and power to all subsystems where selected microcontroller is programmed to control the haptic response. The novelty of the developed electric vehicle control system lies in the use of a haptic technology in the joystick system and vehicle obstacle detection. This allows the vehicle driver to receive perceptible information in the joystick about the approaching obstacles on the joystick. If an obstacle detected by the detection system is not recognized by the operator, the entire control system notifies the driver of the obstacle and simultaneously stops the car in front of the obstacle. If the car is parked in front of an obstacle and the driver tries to move the car (forcing a collision), the system will prevent the joystick from being swung out and the car will not move. In the complexity of vehicle the LiDAR distance sensors have played a crucial role to extract meaningful information from environments and to recognize the surrounding objects using a laser as source. In addition, the experimental tests were conducted to avoid obstacles such as walls or objects of size 15×20 cm and detect collision through the information of the distance to obstacles, joystick angle and vehicle speed from LiDAR, microcontrollers, and sensors. In the conducted experimental tests, the obstacle is placed at distance of 2 m away from the vehicle. If the resistance decreases, it means that the vehicle is approaching the obstacle. The recorded speed of the vehicle will go from 350 mm/s to less than

10 mm/s and the resistance increases to reach 100%. The investigation and study of the realized vehicle in Fig. 1 have been available at the Laboratories of Silesian University, Gliwice, Poland. The paper is organized as follows: the Section 2 introduces the hardware and mechanical architecture of electric vehicle and analog joystick system and Section 3 describes the performed experiments and discusses the obtained results by experimental tests. The Section 4 closes with the conclusions.

2 System Architecture: Mechanical and Electrical System

2.1 Joystick System in the Haptic Feedback Technology

In this paper a haptic feedback was designed and explored to control the information received by force sensors and limit switches inserted in a movable analog joystick controlling an electric vehicle for the obstacle detection. The joystick control system is equipped with four strain gauges located on each side of the knob. In particular the control directions for accelerating and braking of vehicle is allowed by four strain gauges that produce the electrical output signals proportional to the force and in basic directions (UP,DOWN,RIGHT and LEFT) of the joystick. The four strain gauges are mounted on surface of control joystick and are connected to an electronic circuit comprising a differential amplifier for receiving signals. The strain gauges are able to provide electrical signal under various forces and tension applied along the direction of human compression. The force received by the joystick is translated to the referred speed. In this case the maximum tilt of joystick is translated to the maximum speed of the vehicle.

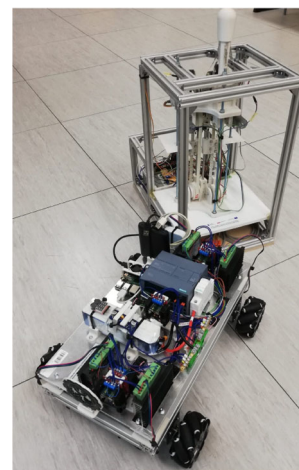
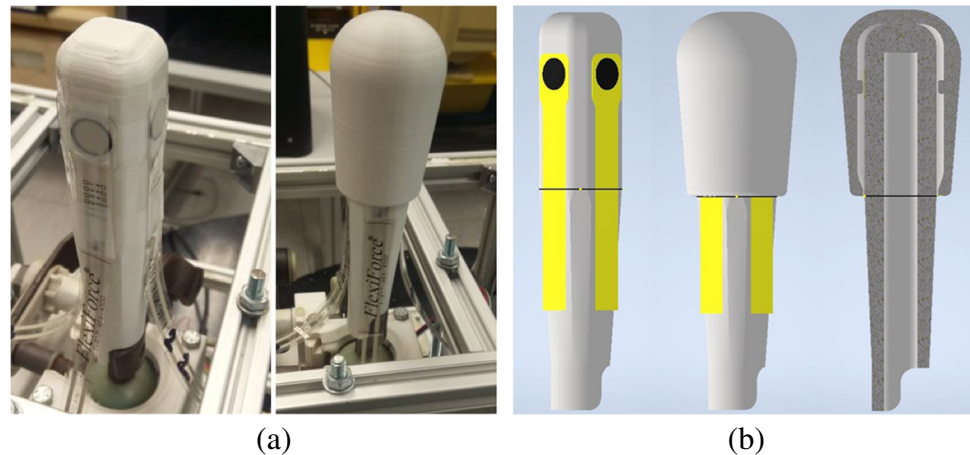


Fig. 1 The developed electric mechanical assembly vehicle, equipped with four Mecanum wheels equipped by external analog joystick

Fig. 2 (a) View of the realized control joysticks with the strain gauges mounted on each side of knob ; (b) External 3D view of joysticks realized in Autodesk Inventor



The four motors are controlled independently. The speed is derived from common Mecanum wheel control algorithms.

The knob consists of two parts: in the inner part the strain gauges are located, and in the outer part, the force is transmitted to the appropriate strain gauge (Fig. 2(a)). The knob was designed taking into account of ergonomics and the reliability of force transmission, more comfortable to use (Fig. 2(b)). The strain gauges are placed at the base and a locking element has been added in order to prevent the knob rotations. The elements are made of PLA material.

Due to the instability of the structure, additional aluminum profiles were stiffened with threaded rods attached to them. The communication between the joystick microcontroller and the PLC was established using the TCP/IP communication protocol. This solution allows to control completely the data transfer between devices and PLC. The cabling was analyzed to connect force sensor and distance sensors. The joystick control system has been analyzed and improved creating code and functions in order to store information received by force sensors and limit switches. When a deflection is detected on the joystick, the vehicle is moving around the checked area. If an obstacle is detected and into collision, the calculating procedure of resistance force generated on the stick is activated. The calculations have included the existing speed of the vehicle and the distance that needs to stop it. In a critical situation, the controller stops the vehicle, regardless of the force applied to the control rod, a few centimeters in front of the obstacle, preventing its damage.

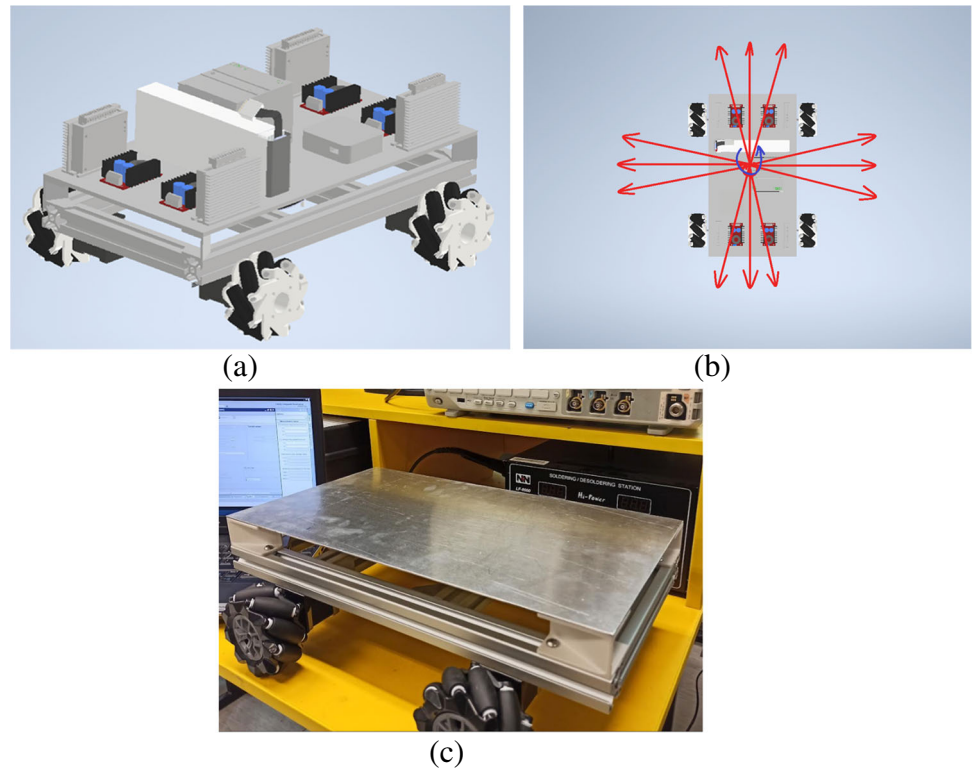
2.2 Remote-controlled Vehicle

The adopted Mecanum-wheel omnidirectional vehicles can move independently in all directions of motion based on a type of holonomic drive with ability to apply the force of the wheel at 45° angle instead of on one of its axes and to gain translational control of the vehicle. The motions can be achieved for various combination of wheel rotation.

The vehicle components include the Mecanum wheels with rollers for treads to drive in all directions and move freely. Since each wheel must be driven independently, four single channel controllers are used. In particular plastic wheels with a diameter of 97 mm with rubberized rollers were selected and the stepper motors to precisely control the position of the vehicle. The vehicle equipped with mecanum omni-directional wheels is powered by Nema 17 stepper motors, whose dedicated drivers convert the signal from the management unit to the appropriate power supply for the engines. Controllers with appropriate power and compatible with Nema 17 motors were selected. Specially Siemens S7-1214C DC / DC / DC PLC controller was selected and designed as compact CPU with ability to communicate efficiently with other devices via an Ethernet network. A fast output signal board is added in order to obtain a PWM signal. A supply unit with appropriate efficiency and Battery Pack: 5000mAh- 6S, 22.2V -100°C were adopted.

The power consumption related to the motion system of electrical vehicle is about 25 W and using the battery of 5000mAh the vehicle moves for more than 3 hours. Due to the different rated voltages of the devices operating in the vehicle six converters with adjustable output voltage were used. Motor controllers and PLC require a power supply of 24V, while the high-speed output motherboard is powered by 5V. Since PLC controller performs the communication with the joystick and the measuring probe, a router is requested for a local wireless network, so connector and Access Point function the power supply of 5V is guaranteed. In order to select the dimensions of the vehicle, a simulation model was created as shown in Fig. 3(a). For the construction of the remote-control vehicle were used Rexroth profiles of 30x30mm, aluminum sheet of 3mm, distance connectors in PLA, bolts and hammer nuts (Fig. 3(c)). Each of four motors is powered by a separate module in turn respectively is supplied by four converters with appropriate current, as well as a PLC controller that gives the control signal. Most of the elements were designed

Fig. 3 (a) Model design of vehicle structure equipped with mecanum omni-directional wheels realized in Autodesk Inventor; (b) Top view of the vehicle equipped with LiDAR sensors able to determine position and distance of obstacles, (c) Omni Mecanum Wheel Robot Car Chassis Platform



and then printed with special mounting brackets, and fixed on sheet metal with allen screws. The connections were realized using wires of the appropriate cross section as shown in Fig. 3. The control unit for the entire vehicle is a PLC and the controller is connected free with Raspberry PI after the configurations and power supply (Fig. 4). If all devices appear in the network, wheel drives and the main control loop of vehicle are activated.

2.3 Laser Distance Sensor

The detection of obstacle is validated experimentally by a LiDAR system implementation which is a remote sensing technology used to find the distance to an object by means a laser as its source guaranteeing high performance, high speed, real-time data acquisition, so it regularly adopted in all the system where the measurement of a distance is required (Fig. 3(b)). LiDAR technology uses the pulse from a laser to collect measurements and create 3D models and maps of objects and environments computing the difference of the wavelength when the laser source is reflected by the obstacle if it is found. A LiDAR system calculates how long it takes for beams of light to hit an object or surface and reflect back to the laser scanner, then the distance is calculated using the velocity of light. LiDAR has the ability to provide a long-range and wide-angle laser scan (360-degree scanning within a range of 6 meters) and 6.5 full revolutions around its axis in one second with a measurement distance range of 15cm-12cm.

The RP LiDAR A1M8 system is controlled using the Raspberry Pi 4 computer.

One of the assumptions of the measuring probe was the independence of operation and the possibility of assembly / disassembly without major interference with the vehicle, therefore a separate Anker PowerCore and 13400mAh power bank were used as power source respectively for the computer and the LiDAR system.

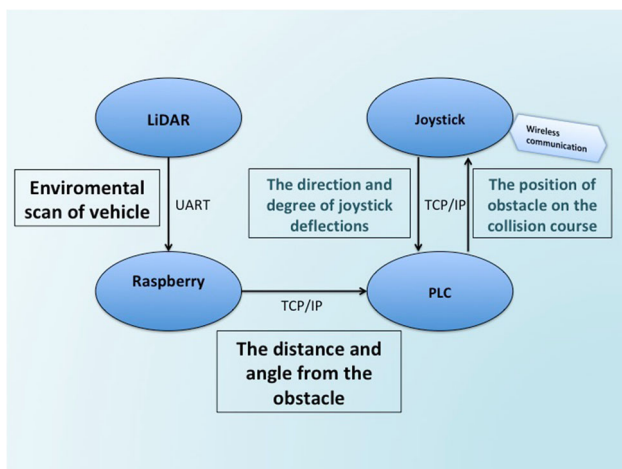


Fig. 4 Dependencies between elements of controlled vehicle and mutual communication

The LiDAR system was located under the plate and on top of the wheels, in order to have a large field of view and, at the same time, a lower minimum height (about 10 cm) for the detection of obstacles. The following procedure of obstacle detection has been applied: based on the tilt of the joystick, the direction of movement is determined, then the original LiDAR range (360-degree) is limited to a range of 45-degree in the direction of movement. Finally the smallest value from the LiDAR sensor (potential obstacle) is determined within this range.

2.4 Wireless Transmission System

The transmission system, selected for electric vehicles, is the wireless technology, however, the Wi-Fi has been used in conjunction with the Ethernet network. The connection between the measuring probe and the PLC turned out to be trouble-free, as the Raspberry Pi 4 has a built-in Wi-Fi module that allows the wireless connection to the router. The PLC controller was connected to the router via an Ethernet cable. To realize the wireless mobile communication of the joystick with the vehicle, Wi-Fi module and ESP32 DevKitC V4 microcontroller has been adopted, beside the considered communication protocol has been the TCP-IP protocol that is easier to implement for microcontroller and PLC (Fig. 5). The communication between the joystick and microcontroller was established using the TCP-IP communication protocol easier to implement for microcontroller. This solution allows to control the completion of data transfer between devices causing delays that are reduced, by limiting the frequency of transferring data from the microcontroller. The Wi-Fi based remote control system of electrical vehicle performs complex operations for the detection and measurement of environment parameters.

The LiDAR has been connected via a dedicated UART → USB converter to the Raspberry Pi 4 computer, the program on which, reads directly the bytes from the UART serial port/connection denoting the distance from the obstacle and the angle by means LiDAR technology. The distance in millimeters is converted to centimeters, and

the angle in degrees. The distances in ASCII code with characters from 0 to 126 are assigned from 0cm to 254cm (each subsequent character indicates an obstacle located 2cm away). Character 127 in ASCII code is reserved for information relative to obstacle outside of the specified range (closer than 16cm or more than 254cm). Distances beyond this range can be ignored, because any obstacle closer than 16cm would mean that within of the range of the vehicle, and any obstacles further than 254cm do not detect the crash with its, taking into consideration the speed of the vehicle.

3 Experimental Work and Results

3.1 The Dependence of the Stick Deflection and the Vehicle Distance

Vehicles with the ability to detect obstacles such as walls or objects of size 15×20 cm were considered an executive factor in analyzing the effects of stick deflection on the vehicle speeds and the obstacle detection on stick response. In Fig. 6 is described the deflection of the joystick that depends on the application of force and the vehicle distance, therefore the joystick deflection directly reflects the set vehicle speed and the bending direction of the joystick corresponds to the moving direction of the vehicle. The effect of obstacle detection on the stick reaction has been determined. The error of the measurement distance between vehicle and obstacle has been estimated automatically by LiDAR sensors and Raspberry Pi 4 system. Obviously, to deflect the stick from its resting position, a minimum pressure force is required and modified according to the relative distance of the obstacle detected, so more pressure is required to deflect the stick or hold it in position as the vehicle moves closer to the obstacles. An additional strength is perceived to the joystick deflection towards the obstacle, and in other directions where no obstacle is detected. The dependence of the required pressure force and the distance close to the obstacle was described by a quadratic function for a more effective haptic effect (Fig. 6).

Fig. 5 Code fragment of communication for Raspberry and ESP. The socket is selected to communicate with the client-server and the server address (Raspberry Network address) is set. Then the connection is established using the bind command. Using the code, the conditions are provided for network connection

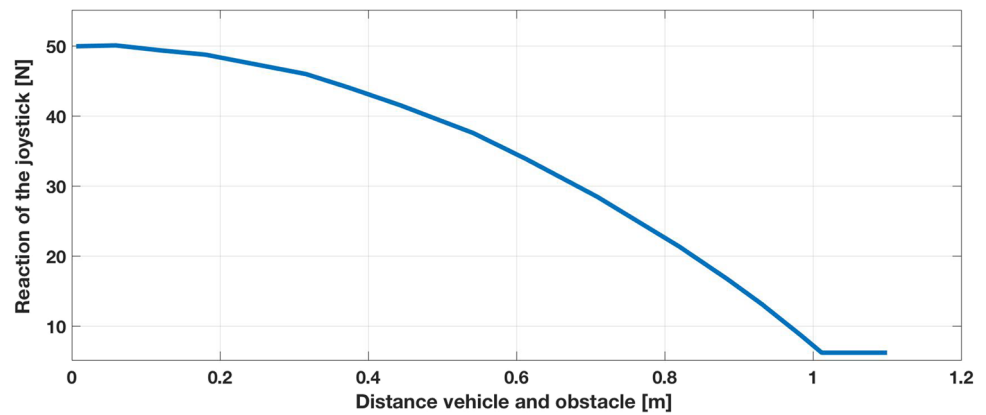
```

17 HOST = '0.0.0.0'
18 PORT = 29998
19
20 with socket.socket(socket.AF_INET, socket.SOCK_STREAM) as s:
21     print("Connected!")
22     s.bind((HOST, PORT))
23     while True:
24         s.listen()
25         conn, addr = s.accept()
26         with conn:
27             print('Connected by', addr)
28
WiFiClient sendData(WiFiClient sc)
{
    if (!c)
    { c = Server.available();
      c.setTimeout(5);
    }

    if(c.connected())
    {

```

Fig. 6 The distance between the vehicle and the detected obstacle with the reaction of the joystick



During the control tests an additional dependence of the required pressure force on the level of stick deflection has been found, squarely, an increasing drag related to the level of stick deflection. The difference between the pressure in the rest position and the pressure at the maximum deflection is 3 N, which greatly facilitates the action of the joystick.

3.2 Position and Force Sensors in the Joystick Sensing System

In the joystick control system, the position and force sensors were adopted.

The SparkFun SEN-08712 contact force sensors (FSR) were selected as Tekscan piezoresistive force sensors with the maximum detectable force of 11 kg. By pressing hard, the resistance increased changes from infinite to 300 kΩ. The sensor itself is thin and flexible with the overall length of 203 mm and his resistance does not change when it is flexed but only when the pressure is applied to the rounded area at the end of the sensor. The Force Sensing Resistors are used in the vehicle to measure and detect the magnitude of the user's force on joystick. Four sensors are located on the surface of the joystick which exhibit a decrease in resistance with an increase in the force applied on it. The measurement system has provided the read of the variation of the variable resistance. In such system the FSRs sensors are connected to the microcontroller's ADC input and the signals are amplified and enhanced using the operational amplifier MCP6002, in which the appropriate pins were used as power supply for the system with the 3.3 V logic, on which the ESP works (Fig. 7).

Several tests were then carried out to determine the characteristics and to check the repeatability of the sensor. The increased pressure force was measured using a laboratory dynamometer located on the drill handle with tests that confirmed the repeatability and reproducibility of the sensors and that demonstrate how their resistances

read on the microcontroller, vary linearly with the applied pressure force. In the Fig. 8, the maximum value of the read pressure force is 60 N, but this value can be changed by adjusting of the potentiometer of the measurement system as appropriate and the maximum value of force is 50 N (Fig. 8). As a result of the applied contact force, the SparkFun sensor varies the resistance in a non-linear manner. Therefore, a measuring circuit suggested by the sensor manufacturer gives a voltage signal proportional to the applied force. Figure 8 shows the relationship between the force applied to the sensors and the value read by the ADC. These tests were carried out to determine repeatability of sensors.

3.3 Experimental Tests Conducted for the Obstacle Detection of Vehicle

The experiments were carried out to avoid the obstacles and detect the collision with LiDAR data. The obstacle was located 2 m from the vehicle, then, the joystick was swung in such a way that the vehicle was directed towards the obstacle so the dependence of the required force variation on distance from the obstacle was verified. Multiple experimental tests were conducted for both forward and lateral direction. The omnidirectional vehicle equipped with four Mecanum wheel provides accuracy in the centimeter range. The results have confirmed the effectiveness and robustness in presence of uncertainties and unknown external obstacles using the LiDAR and microcontroller. The parameters recorded every 100 ms during the experiment include:

- four sets of current speed motor related to each Omnidirectional Mecanum wheel
- angle set by joystick (Fig. 9)
- speed of the vehicle expressed in mm/s
- resistance Re that is the inverse distance of vehicle to obstacle expressed in %

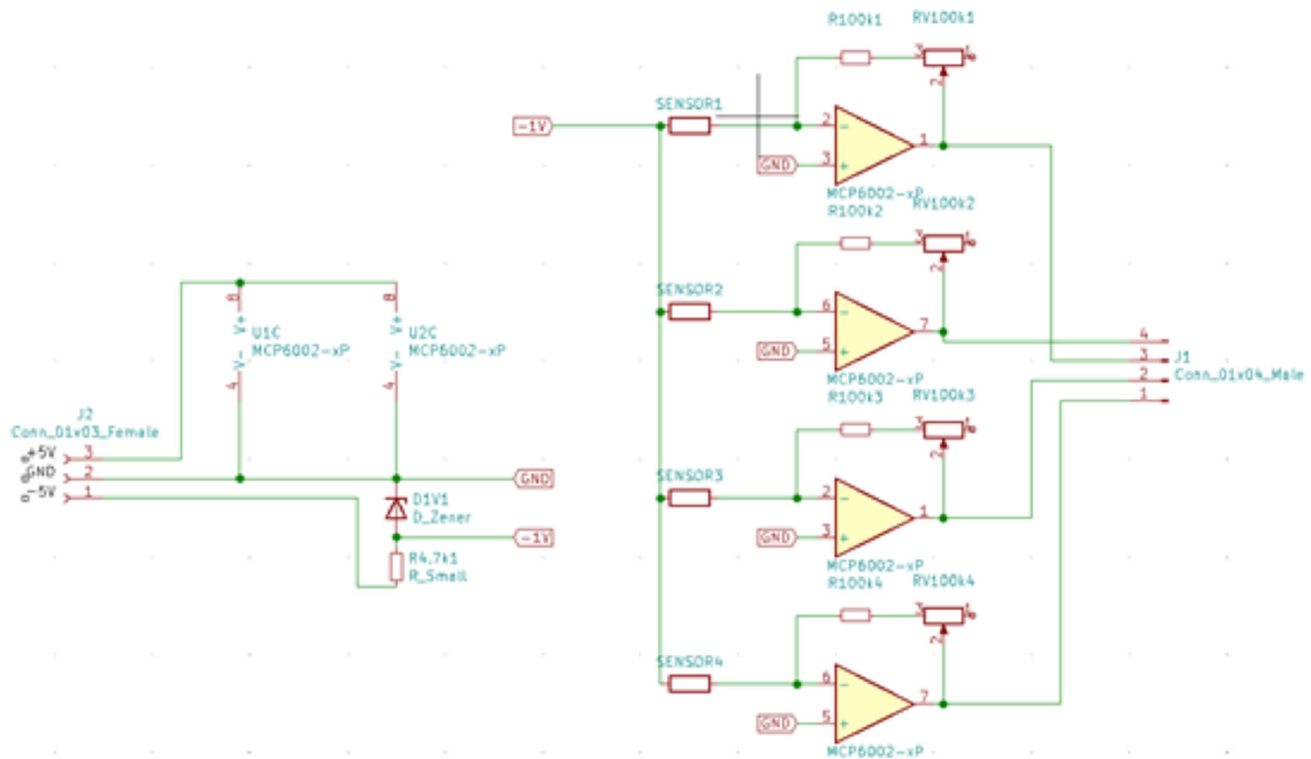


Fig. 7 Schematic of measuring system for pressure force associate to the four force sensing resistors (FSRs). The four sensors are interfaced to the microcontroller and connected to the MCP6002 operational

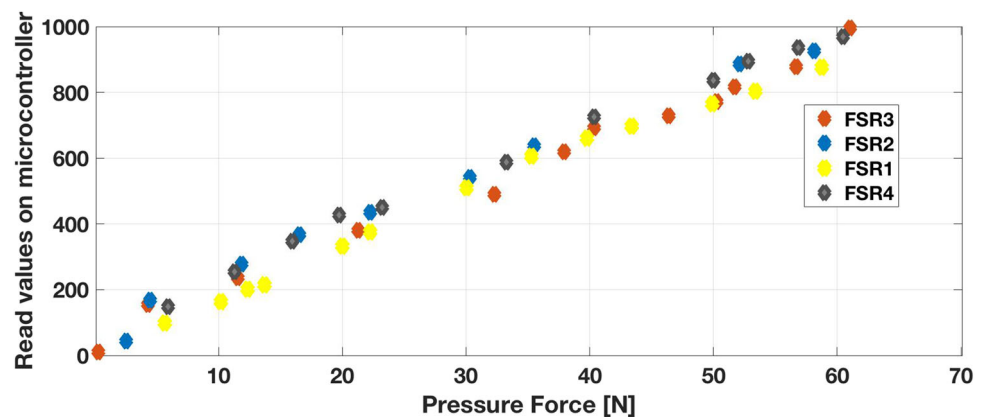
amplifier to detect the changing of the resistance in linear relation with force applied by the user on joystick. The output of amplifier is voltage signal in linear relation with force applied by the user on joystick

- four steering signals of motion in UP, DOWN, RIGHT and LEFT direction obtained by force sensor/ ADC and expressed in V
- distance from the obstacle expressed in *mm*

Each measurement was carried out as a function of the force applied by the user and detected by the sensor, in particular, the input force is the human pressure force applied on the joystick. The LiDAR sensor measures the distance from the obstacle to the vehicle, and the microcontroller records sensors and motors signals, the distance between the vehicle and obstacle, and the

movement of the joystick. The microcontroller can move the joystick in the UP, DOWN, RIGHT and LEFT direction so that the vehicle will move forward and backwards or left and right direction (Fig. 10). If the vehicle is closed to the obstacle, the microcontroller can arrest the vehicle and the speed of motor can slow down such that the joystick switches to zero position until the joystick is blocked and the vehicle comes to a stop. The speed depends on the current of motor as well and it is proportional to the distance. The workflow algorithm to avoid obstacles along the travel path is presented in the Fig. 11 The initial stage includes the pre-processing and initialization of the drives. At this

Fig. 8 The pressure force characteristics of the four FSRs for which the values of force are read by the microcontroller



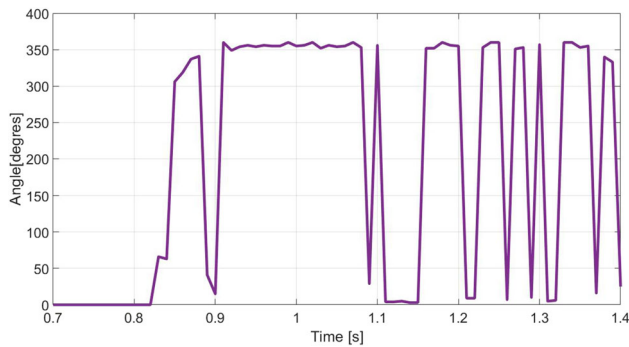
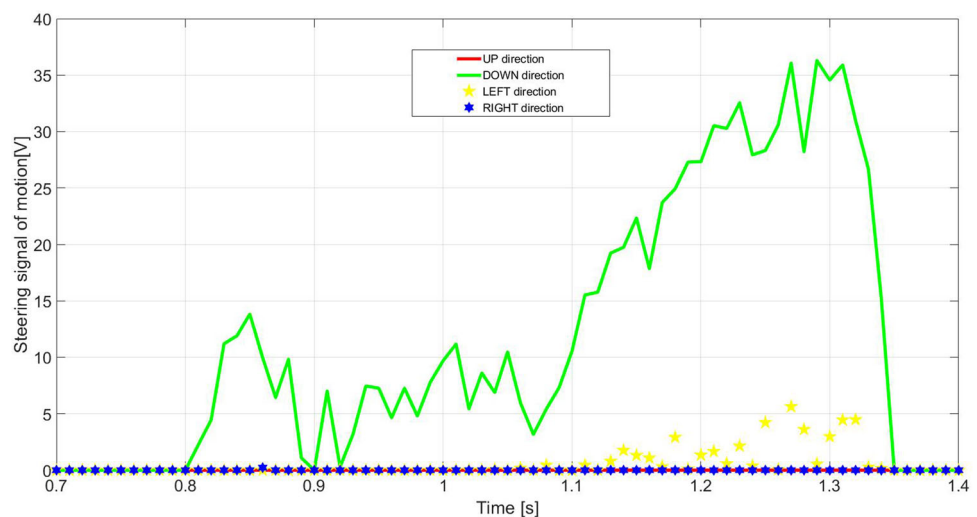


Fig. 9 Angles controlled by joystick for a given deflection of vehicle

stage, the directions of the joystick, in particular the four movement steering signals are read by microcontroller and it is possible to estimate the presence of the obstacle. Next, the calculation of the resistance and distance from obstacle are performed. Based on the collected data from the above-mentioned procedures and the LiDAR sensors, the vehicle stop to avoid the collision or moves.. The Fig. 12 shows the dependence of resistance (or more properly the inverse of scaled distance) at a given vehicle speed by considering the four movement steering signals in the UP, DOWN, RIGHT and LEFT directions of the joystick (Fig. 10), however jumps and jerks are presented in the collected data caused by inaccuracy of the surroundings scanner. The number of samples, in our case, is generated every 100ms. The resistance is an information from LiDAR and it depends on distance between the vehicle and the obstacle, it is calculated dynamically according to the vehicle speed and the distance from the obstacle. The resistance Re is expressed by formula of scaling as in Eq. 1

$$Re = \frac{d \cdot \text{Max}.Re}{\text{Max}.d} \quad (1)$$

Fig. 10 Four steering signals of motion of joystick/vehicle in UP, DOWN, RIGHT and LEFT direction

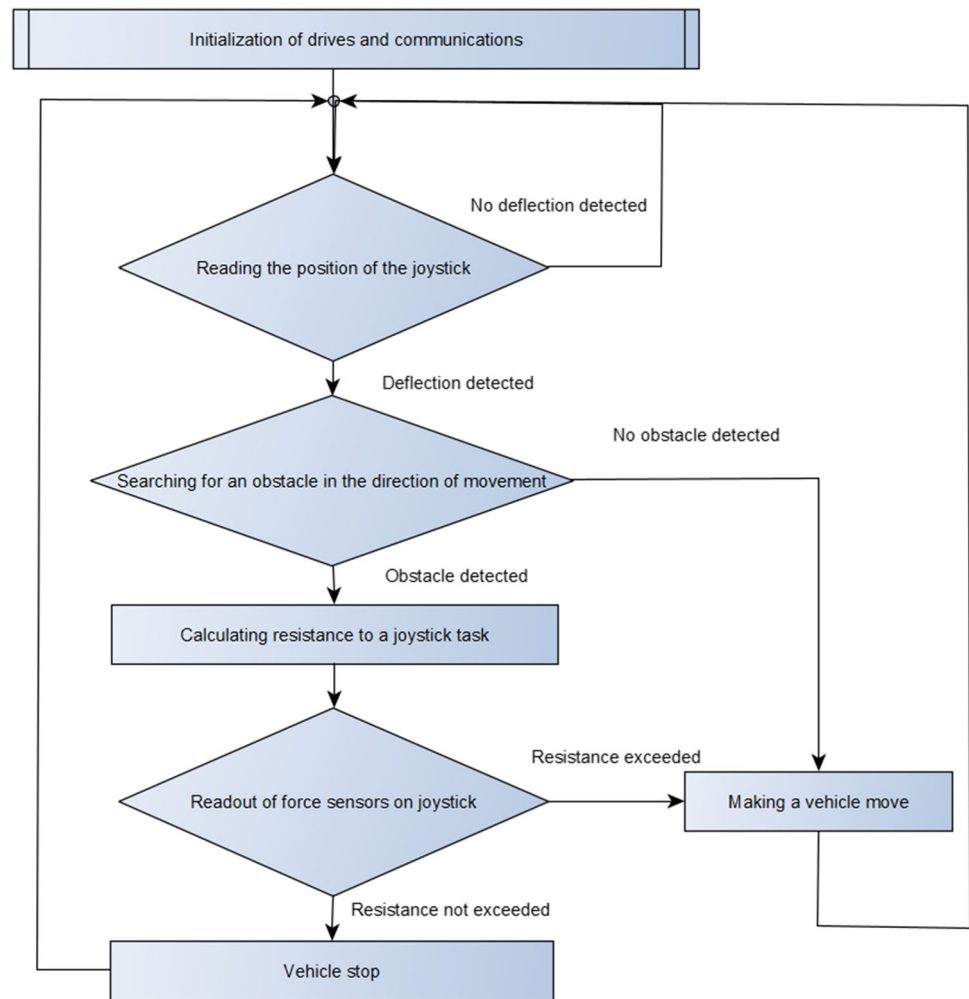


where $\text{Max}.Re$ is the maximum value of the resistance equal to 100, $\text{Max}.d$ is the maximum value of distance (1000 mm), and d is the distance measured from LiDAR sensor expressed in mm. If the resistance decreases, it means that the vehicle is getting close to an obstacle, instead if there is no resistance, this corresponds to 100%, which means that there are no obstacles and the vehicle is driving in different directions. A full resistance value, corresponding to a maximum value of 0 % in this case, means that the obstacle is closed to the vehicle, the joystick is blocked, and the speed of vehicle slows down, as shown in the Fig. 12 in an interval of 7 seconds, in this case to maintain the vehicle speed, it is necessary to increase the pressure applied on the control stick. The dependence between the speed of vehicle and the resistance is shown in Fig. 12.

4 Conclusions

In this research activity, an implemented vehicle is provided by four Mecanum wheels on the corners of a rectangular base with omnidirectional motion equipped by Raspberry Pi and a LiDAR module for avoiding obstacles and navigating in an unknown environment. The interface with the vehicle's mechanical and electrical devices is performed via remote control with haptic feedback using an analog joystick that guarantees the driving movements. The haptic feedback improves the operator's awareness of the area surrounding the vehicle and assists the driver in obstacle avoidance. Therefore, the conducted experimental results explain the dependence of the deflection of the pressure applied to the joystick and the vehicle distance to the obstacle described by a quadratic function for a better tactile effect. Furthermore, experiments were performed to avoid the obstacle and detect the collision by information from LiDAR, microcontroller

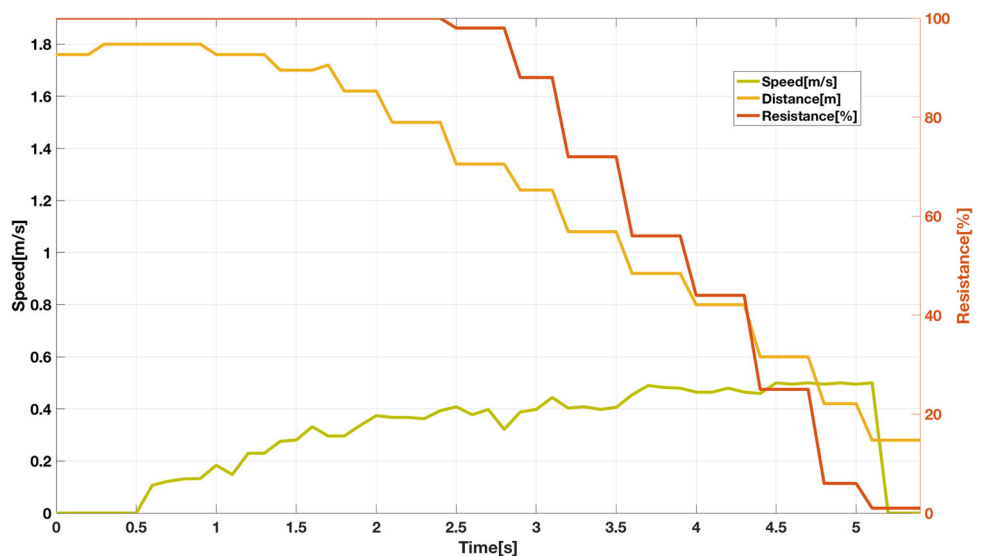
Fig. 11 Basic concept of the obstacle avoidance executed using the calculating resistance of joystick and the force sensors



and sensors such as the distance to the obstacle, the joystick angle and the speed of vehicle. In such case the obstacle such as walls or objects of size 15×20 cm is placed at

distance of 2 m away from the vehicle. If the resistance decreases, it means that the vehicle is approaching the obstacle. The recorded speed of vehicle passes from 500

Fig. 12 The vehicle speed and the resistance are related to the distance between the vehicle and obstacle. The obstacle was placed at distance of 2 m away from the vehicle. If the resistance decreases, it means that the vehicle is approaching an obstacle. If the distance between obstacle and vehicle is over than 1.4 m the resistance is 100%. If the distance is in the range 0.3–1.4 m, the resistance is proportional to the distance. Finally if the distance is less than 0.3 m, the resistance is equal to 0 and the vehicle has to stop immediately



mm/s to value less than 10 mm/s, on the contrast, the resistance increases to reach 100%.

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Data Availability The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval This article does not contain any studies with human participants or animals performed by any of the authors.

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